



Nationalism meets machine heuristics: Investigating the effect of AI's "nationality" on the perceived credibility of AIGC

Junyi Chen^a, Weixi Zeng^b, Yi Mou^{a,*}

^a School of Media and Communication, Shanghai Jiao Tong University, 800 Dongchuan Road, Minhang District, Shanghai, 200240, China

^b School of Journalism and Communication, Nanjing University, 163 Xianlin Road, Qixia District, Nanjing, 210023, China

ARTICLE INFO

Keywords:

Politicized technology
Credibility
AIGC
Machine heuristic
Nationalism
Large language model

ABSTRACT

Against the backdrop of politicizing science and technology, it has nevertheless become a common practice to employ AI tools in political communication. This study delves into the tension between the supposedly high credibility of AI-generated content (AIGC) with its perceived objectivity and the potentially devastating impact of politicized tools. Specifically, we investigate the disparities in the perceived credibility of AIGC, stemming from various source types and nationalities. Using an online experiment conducted in China, we found that Chinese AI sources and human-AI hybrid sources with consistent Chinese national identity were perceived as more credible than their foreign (American) counterparts, whose effects were fully mediated by machine heuristics. In contrast, less credibility was directly attributed to human-AI hybrid sources with inconsistent human-AI nationalities (e.g., a Chinese journalist using American AI). Furthermore, although foreign sources were generally seen as less credible, the presence of AI in the source moderated this negative effect. These findings indicated users' perceptions of the political dimensions of AI technology.

1. Introduction

Along with the recent advancement in generative artificial intelligence (AI), automation of content production has profoundly influenced the existing media landscape. An increasing number of news articles are being written by AI (Diakopoulos, 2019), social media is flooded with bots (Bessi & Ferrara, 2016), and AI-generated content (AIGC) has gained widespread attention due to the rise of large language models (Lopez, 2022). This trend could complicate the already divisive and polarized online discussions (Hart et al., 2020). For example, scholars are debating whether AI authors should be disclosed or if human-AI co-authorship should be introduced, which may trigger different political effects. Optimistic researchers highlight the potential of generative AI as a more credible information source than humans, especially in the political domain, due to its perceived lower levels of bias (Chen et al., 2024), higher levels of credibility (Liu et al., 2018), and ability to induce more willingness to disclose privacy (Sundar & Kim, 2019), inspired by the concept "machine heuristic" (Sundar & Kim, 2019). This indicates that people may hold a stereotype of machine sources as being objective and nonpartisan, assigning higher levels of credibility and lower levels of motivated reasoning (Chung et al., 2023). However, AIGC is implicitly

biased by factors such as race, gender, and age from its training data (Sun et al., 2023), and AI can trigger radicalism and polarization often without users' awareness (Burton, 2023). AI sources are increasingly used in communication practices to reduce political biases and motivated reasoning (Sætra, 2020), and its effect demands further examination.

Technology is usually oversimplified as apolitical (Verbeek, 2017). Previous research on AI and source credibility frequently considered it a stateless communicator, assuming it originates from its home country by default [e.g., Banas et al., 2022; Verbeek, 2017]. Indeed, in contemporary times, most technologies are shared across national borders. However, as a result of global technological competition, concerns have arisen about national policies on emerging technologies, which may impact their perceived credibility. For example, in Russia, local social networking sites Odnoklassniki and VKontakte are more popular than Facebook and Twitter due to "nationalist considerations and concerns over national sovereignty" [Jang et al., 2023, p.337]. Similarly, in China, perceptions of genetic modification are politicized, driven by the suspicions that American technology will genetically alter the Chinese genes. Therefore, when AI's nationality meets the "machine heuristic," will the machine sources still be considered neutral and objective? To

* Corresponding author.

E-mail address: yimou@sjtu.edu.cn (Y. Mou).

<https://doi.org/10.1016/j.techsoc.2025.103032>

Received 3 April 2024; Received in revised form 21 July 2025; Accepted 31 July 2025

Available online 5 August 2025

0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

what extent does the machine heuristic continue to exert an impact on individuals' perceptions of credibility?

Against the backdrop of politicizing science and technology (Brown, 2015), this study delves into the tension between AIGC's supposedly high credibility as AI sources and the potentially devastating impact of politicized tools. Specifically, we investigate the disparities in the perceived credibility of AIGC, stemming from various source types and nationalities, through an online experiment. Furthermore, this paper explores the implications when both human and AI sources are mixed. Given that prior research has scarcely addressed the role of AI's "nationality" in communicating political messages, this study aims to fill this gap by adopting a technological and political lens.

2. Literature review

2.1. Perceived credibility of human, AI, and Human-AI hybrid

As human-machine interaction evolves, machines have moved beyond merely mediating communication to generating complex messages independently (Hancock et al., 2020). With the increasing sophistication of AIGC, people can barely distinguish whether the content is generated by AI or human (Jakesch et al., 2023). Since AI has progressed from providing simple factual content to generating controversial information (Guzman, 2019a), it is essential to examine the credibility of AIGC.

Humans hold a wide range of preconceptions about AIGC [e.g., Guzman, 2019a], even though current AI is already incomparably closer to humans in terms of content generation. The labeling of AI sources may influence audience attitudes. Several studies have shown that AI is often considered more objective, accurate and nonpartisan than human sources (Cloudy et al., 2021, 2022; Moon et al., 2023). However, a large amount of different evidence exists at the same time. They find no difference in credibility between AI and human sources (Kaa & Krahmer, 2014; Xu et al., 2020). Another meta-analysis notes that AI is as persuasive as humans in terms of overall persuasive outcomes (Huang & Wang, 2023). There is also experimental evidence that labeling a Chat-GPT-generated scientific article with human authors will result in rise in anthropomorphism, intelligence, and a slight increase in perceived credibility (Lermann et al., 2024). Notably, however, those studies mainly focus on topics like sports, finance, science, and arts rather than controversial political issues.

Theoretically, Sundar's Modality-Agency-Interactivity-Navigability (MAIN) model argues that people have a predisposition that machines are more impartial and neutral than humans, known as the machine heuristic (Sundar, 2008). It can be viewed as a heuristic information processing that is activated by exposure to a machine source (including AI). Guzman found that people usually perceive themselves as directly interacting with AI rather than with a programmer behind it (Guzman, 2019b). It indicates that the public generally does not perceive the human factor behind the technology. In other words, technology itself is the cue that triggers heuristic thinking.

Heuristic thinking, in turn, is widely used as a mental shortcut when people are unwilling or unable to evaluate information exhaustively, especially in the context of information overload and cognitive overload (Chaiken, 1987). Evidence has pointed out that this unconscious processing mode may be more influential than conscious judgment (Xu et al., 2022). According to the classic heuristic-systematic model (Chen & Chaiken, 1999), under heuristic information processing, the credibility of the source influences people's evaluation and judgment of the information. Therefore, individuals may believe that the source and the content generated by machines are objective and unbiased due to the positive impression arising from machine heuristics. To put it differently, people will associate machines with some "machine-like" qualities, such as objectivity, impartiality, accuracy, absence of error, unemotionality, indomitability, and reliability (Sundar, 2008), reducing perceived bias and politically motivated information processing (Hong

et al., 2025), leading to greater perceived credibility.

Some researchers believe that for socio-political topics, AI-authored news induced less credibility than human-authored news due to AI's limitation in deep analysis, metaphor comprehension, and contextual awareness (Wang & Huang, 2024). But in contrast, in today's political information, people weigh their pre-existing beliefs much more heavily than facts and truth (Stanley et al., 2020). Differences in preconceptions have a greater impact on credibility than factual arguments (Gollwitzer et al., 2020). Thus, the perception that AI is free from ideological bias is more likely to activate machine heuristics, making it more credible than human sources.

Beyond the human versus AI dualism, future trends may embrace the form of a "human-machine marriage." For instance, in journalism, AI may handle fact retrieval and initial draft writing, while human journalists focus on refining and deepening the analysis (Graefe, 2016). This trend has become increasingly evident. With humans having the final say, the transformative power of AI has been increasingly applied to content production. Some news organizations have begun to take AI and humans as co-authors (Luke, 2023). However, the credibility of the human-AI hybrid source has not been much studied, in par with its solo human and solo AI counterparts. It is often assumed that the human-AI hybrid source would combine the strengths of both humans and AI. The content created collaboratively by humans and AI would stimulate machine heuristics in readers, reducing perceived bias, while being more relatable to readers because of human sources through the principle of similarity attraction (Byrne, 1997). Indeed, research suggests the hybrid source of political information was significantly attributed to more credibility for American conservatives (Waddell, 2019). In a laboratory study where humans and ChatGPT collaborated on a writing task, participants reported a relatively high level of cognitive trust and satisfaction (Luther, Kimmerle, & Cress, 2024). Yet, opposite evidence suggests that the credibility of the hybrid source lies between that of a solo human and AI source, with no significant differences between the two (Tandoc et al., 2020; Wölker & Powell, 2021). One recent study also explores the role of authorship in scientific journalism, suggesting that there was no difference in credibility between human-AI hybrid sources and solo AI authors. However, after adding a disclaimer to the scientific article, hybrid sources were more credible than solo AI sources (Lermann et al., 2025). This further calls for detailed research on human-AI hybrid sources in different conditions. To examine the differences in perceived credibility among those three types of sources, H1, H2 and RQ1 were formulated:

H1. AI is perceived as more credible than humans as the source of political information.

H2. Perceived machine heuristic mediates the effect of source type (human vs. AI) on perceived credibility.

RQ1. Is there any difference in the perceived credibility among a human-AI hybrid source, a solo human, and a solo AI as the source of political information? If so, does the machine heuristic play a mediating role?

2.2. The impact of nationality on source credibility

The development of communication technology has allowed information to cross national boundaries, making the country of the source increasingly important. Research has suggested that the nationality of human sources affects communication effects, often placing foreign sources at a disadvantage compared to domestic ones (Dragojlovic, 2015). For example, American media are less credible in the Arab region due to religious reasons (Golan & Kiousis, 2010). Another study found that the greater the number of opposing views expressed by foreign sources, the more support there was for military action among highly educated Democrats in the United States during the Iraq War [Lermann et al., 2025, p.843]. Thus, a national perspective remains in global

communication. One possible explanation could be the “rally-round-the-flag” phenomenon, whereby the public tends to be more receptive to patriotic discourse during political conflicts, leading to increased trust in their own country’s government and media (Bryanov et al., 2023).

Social identity theory explains the effect of the source’s nationality on credibility at the individual psychological level. It suggests that certain social group factors, such as sex, age, and nationality, can help individuals build a sense of self-identity by relating themselves to these factors. Individuals often identify with the groups they belong to and may prefer those within their own group (Tajfel & Turner, 2004). One study found that attempts to change people’s preconceptions were often ineffective when tied to their national identity (Mann & Schleifer, 2020). It was supported by various classic studies in political psychology [e.g., Golan and Kioussis, 2010; Zhu et al., 2022]. People often lack full information and do not rationally analyze complex political and social issues. Instead, they rely on the psychological shortcut of identification to form political preferences and trust. Through the identification mechanism, people are more likely to perceive higher levels of credibility in internal group members, while exhibiting motivated skepticism toward external group members (Brewer & Chen, 2007). Thus, in the case of political information, people will have less trust in foreign sources.

The computers are social actors (CASA) paradigm and media equation theory suggest that people tend to mindlessly treat new media like real people and apply a range of social norms and scripts from interpersonal interactions (Reeves & Nass, 1996). Past research has shown that people project human traits and stereotypes onto machines. For example, robots with female characteristics are seen as more suitable for care work than male robots (Tay et al., 2014), and AI algorithms labeled younger are seen as more competent (Luther, Duzel, et al., 2024). Likewise, foreign AI sources may maintain a lower level of trustworthiness when people realize that the machines are from other competing countries, similar to their attitudes toward people from those countries. The concept of nationalism also helps to provide an explanation. Scholars have introduced novel concepts such as vaccine nationalism (Barceló et al., 2022), digital nationalism (Lu, 2024), and space nationalism (Molle, 2023) to describe public attitudes that favor domestic technologies. One study involving over 15,000 participants across 14 countries revealed that domestically developed vaccines were more popular than American ones, even in countries like France, Singapore, and South Korea that cannot produce high-quality vaccines (Barceló et al., 2022). Moreover, as a highly politicized technology, AI, 5G and other information technologies are seen as highly politicized and contribute to international competition, fostering strategic mistrust of China globally (Kim et al., 2020; Zhang, 2024). This evidence also suggests that the public may be more inclined to accept political messages created by domestic AI. However, the machine heuristic provides an alternative explanation: since the audience tends to believe that AI is objective and unbiased, they may not engage in motivated reasoning, even if the AI source comes from an external group.

In this study, we define AI’s nationality as the country in which its parent company is based. For example, although China and India are major manufacturing hubs for iPhones, people tend to perceive iPhones as American technology. This is because its company, Apple, is headquartered in the United States, whose technology products are shaped by the American culture (Campbell & La Pastina, 2010). As a result, in this study, AI’s nationality is based on the country of origin of its parent company, consistent with the most straightforward public perception of AI’s nationality.

In light of this discussion, H3 and RQ2 are proposed:

H3. a) Domestic human sources and b) domestic AI sources are perceived as more credible than foreign sources as the source of political information.

RQ2. Is there an interaction effect of source type (human vs. AI vs.

human-AI hybrid) and source’s nationality (domestic vs. foreign) on perceived credibility?

2.3. National origins of Human-AI hybrid sources

While the United States currently leads in AI technology, other countries or regions—such as China, the European Union, South Korea, Japan, Russia, and India—are also actively advancing their technology. This has sparked a global technological race (Feijóo et al., 2020). Along with AI products being applied transnationally, communicators will likely generate information with the help of AI technologies from various countries. In other words, when humans and AI act collaboratively as a hybrid source, the countries of humans and AI may be inconsistent. What influence do the nationalities of humans and AI have in a hybrid source?

People usually have certain expectations of communicators, and their evaluations will vary depending on whether these expectations are met, over-met (positive expectation violation), or not met (negative expectation violation) (Burgoon, 2015). One qualitative study suggests that it is a common strategy people use to assess the credibility of AIGC (Ou et al., 2024). For example, people would expect machines to be less biased than humans (Sundar, 2008). As a result, their source credibility will decrease significantly due to negative expectation violation when strong emotional (Tandoc et al., 2020) or evaluative (Henestrosa et al., 2023) overtones are discovered in automated news.

However, researchers further found that this negative expectation violation does not occur in news labeled as co-authored by humans and machines, as no significant credibility difference existed regardless of whether the hybrid sources provide objective facts or emotional opinions. This result is consistent with human sources (Tandoc et al., 2020). Therefore, it can be inferred that readers do not have machine-like expectations of objectivity in response to hybrid sources. Instead, people may tolerate hybrid sources producing emotional content. Another study also found that adopting machines as co-authors in credible newsrooms, like the New York Times, further enhances source credibility. However, the credibility will decrease with a less credible newsroom (Liu & Wei, 2019). This decline occurs because machines are commonly assumed to be used in specific social scenarios. Therefore, accountability for the content typically rests with humans rather than AI. It is inferred that users tend to have expectations based on humans rather than AI.

In times of political conflict, people may prefer their compatriots to use domestically developed technology, while foreign technology may be met with suspicion. For instance, China and the United States are attempting to reduce their reliance on each other’s key technologies. Americans are boycotting the use of Chinese-made Huawei mobile phones, which are perceived as contributing to the “China threat theory” (Tang, 2020). Meanwhile, China strives for a domestic large language model to achieve “technological sovereignty” (Lian et al., 2023). There may be an expectation of consistency between human’s nationality and AI’s nationality in the hybrid sources. Once a human uses AI from another country to publish a political message, it may trigger negative expectation violations that result in decreased credibility. The evaluation of the credibility of human-AI hybrid sources remains contingent upon the social identity of the source in question. In addition, research has shown that users prefer consistent sources over sources with excessive disparities, as consistency generates less cognitive load (Jung et al., 2015). For example, users report more credibility when interacting with a single chatbot than with multiple chatbots (Tan & Liew, 2022). When humans and AI from different countries collaborate on political messages, it may lead to excessive cognitive load. Therefore, we propose H4 and RQ3 to investigate this issue:

H4. A domestic human-domestic AI hybrid source is perceived to be more credible than a foreign human-foreign AI hybrid source for political information.

RQ3. In the case of human-AI hybrid sources, is there any interaction effect between human's nationality and AI's nationality on their perceived credibility?

3. Method

3.1. Procedure

Given the significant influence of US-Sino rivalry on current global affairs, the Chinese public provides a suitable sample to explore the credibility difference in political information generated by different sources. An online experiment with 2 (source's nationality: China vs. USA) $\times$ 3 (source type: human vs. AI vs. human-AI hybrid) factorial design was conducted after getting approval from the internal review board of Shanghai Jiao Tong University (no. H20240469I). Two more condition groups were added as the "nationalities" of human and AI may be different in hybrid sources. Together, those eight groups comprised four single sources (American AI, American human, Chinese AI, Chinese human) and four hybrid sources (American human-American AI, American human-Chinese AI, Chinese human-American AI, Chinese human-Chinese AI). For the Chinese participants, American sources were deemed foreign, whereas Chinese sources were considered domestic.

After providing the demographic information, each participant completed a pre-test to report their pre-existing views on the scenario used in the stimuli. Participants were then randomly assigned to one of the eight groups, completed a manipulation check, and filled out a post-test questionnaire. The participants were debriefed and thanked after they finished answering the questions.

3.2. Sample

A prior power analysis was performed using G*Power 3.1 (Faul et al., 2009). We used the effect size $d = 0.15$ adopted from a meta-analytic study on the impact of machine authorship on news audience perceptions (Wang & Huang, 2024). The results showed that a minimum sample size of 343 was required for 80 % power.

A total of 903 participants were recruited from a large public university in East China, and each received RMB five Yuan for their participation. After we excluded samples that failed the manipulation check and spent less than 60 s in our questionnaire, the final sample totaled 544, with a recovery rate of 60.24 %. Previous research suggested that participants often fail to recall the source of the information they read (Cook et al., 2006; Tandoc et al., 2020), so a recovery rate of around 60 % was acceptable.

The sample consisted of 60.85 % males ($n = 331$) and 39.15 % females ($n = 213$). 90.07 % of the sample fell within the age range of 18–30 years, with 33.27 % aged 18–25 years and 56.80 % aged 25–30 years. In terms of the education level, 63.05 % ($n = 343$) reported they had a Bachelor's degree, 14.71 % ($n = 80$) obtained a postgraduate degree, and 22.24 % ($n = 121$) lower than a Bachelor's Degree.

3.3. Stimulus

The experimental stimulus was fictitious. It presented a story about American officials planning to block the export of Chinese quantum computers, and they advocated for the trade ban because of the data security threat to the United States. The selection of this topic as a stimulus was based on several reasons. First, the topic was politically contentious, reflecting the recently escalating confrontation between the United States and China in the high-tech domain. Second, this topic was timelier compared to some traditional China-US controversial issues. Third, despite concerns about the validity of single experimental material, previous studies on AI sources and credibility have found no significant difference between multiple political topics [e.g., Gollwitzer et al., 2020; Molle, 2023]. Several studies have opted to use a single

experimental story and were considered acceptable (Cloudy et al., 2021; Waddell, 2018). Each stimulus contains 550 words, a typical length for an online news brief.

The human sources used in the experiment were identified by nationality and labeled as "He Xiao" for domestic and "Alex Mitchell" for foreign sources. These names were gender-neutral and common in their respective cultures, to avoid gender bias interfering with the results of the experiment. The human sources were authenticated by their real names and Internet Protocol addresses, with the domestic source located in Shanghai, China and the foreign source in California, the United States. Real-name verification and address exposure are mandated by recent Chinese online regulations for social media platforms (Ministry of Justice of the, 2022). This practice served to authenticate the identity of human authors without causing suspicion. The domestic and foreign AI sources were signed as "Zhiwen 4.0" and "ThoughtFlow 4.0," respectively. The names of the two large language models were also fictitious to avoid the participants' preoccupation toward a real AI product. When introducing the AI sources, a note was added to show that it was a "Chinese AI large language model" or "GPT in California, USA." The human-AI hybrid source combined information from both human and AI sources, with a clear indication that the content was generated by the human source using the large language model. Additionally, details about the number of reads, likes, comments, and mobile phone brands were concealed to avoid confounding effects.

3.4. Manipulation check

The manipulation check was conducted by asking participants to recall the source they had just been exposed to. Participants were required to make the correct choice out of the eight source options. A total of 359 participants failed in the manipulation check and thus were excluded from subsequent data analyses. No significant differences were observed between these excluded individuals and the final sample regarding the experimental conditions ($X^2(7) = 11.099, p = .134$), or nationalism, our core control variable ($t = 0.390, p = .700$).

3.5. Measures

3.5.1. Dependent variables

Source credibility was measured by seven items from Meyer (Meyer, 1988). The participants were asked to indicate the extent they perceived the source to be "fair and objective," "unbiased/neutral," "complete," "trustworthy," "professional," "accurate," and "generally credible." Participants rated these items on a 7-point Likert-type scale, ranging from 1 (completely disagree) to 7 (completely agree) ($M = 4.77, SD = 1.41, \alpha = .96$).

Message credibility was measured with three items (Appelman & Sundar, 2016). Participants were asked to indicate the extent to which the article they read was "accurate," "authentic," and "believable" on a 7-point Likert-type scale, ranging from 1 (completely disagree) to 7 (completely agree) ($M = 4.83, SD = 1.47, \alpha = .93$).

3.5.2. Mediating and control variables

Machine heuristic was measured with a seven-item semantic differential scale. Respondents were asked to indicate the extent to which the source they just encountered was objective, orderly, efficient, accurate, unemotional, logical, and reliable ($M = 4.84, SD = 1.38, \alpha = .95$) (Banks et al., 2021; Cloudy et al., 2021).

Relative advancement of Chinese AI relative to American AI refers to the perceived difference in competence between Chinese and American large language models, which may influence credibility. It was measured using a single seven-point, Likert-style item: "In the field of generative AI, compared to the United States, do you think that China is behind or ahead?" (1 = completely behind to 7 = completely ahead) ($M = 3.80, SD = 1.35$).

Pre-existing view in the selected topic was gauged by the pre-test

question asking about participants' attitudes toward the United States frequently banning the import and export of Chinese high technology in recent years, as their initial view could influence perceived credibility. Responses were rated on a 7-point Likert-type scale ranging from 1 (completely disagree) to 7 (completely agree) ($M = 2.18$, $SD = 1.45$).

Nationalism includes a sense of national superiority towards their ethnic group and the devaluation of other ethnic groups, which is often considered to contain a motive for rejecting foreign countries (Mummendey et al., 2001). Individuals' nationalism can impact attitudes towards foreign AI. In this study, Chinese nationalism was measured using five items: "When China has a serious conflict of interest with other nations, we can use any means to defend its own interests," "Chinese people are one of the smartest and most diligent peoples in the world," "Chinese culture/traditional value is superior to other cultures/civilizations in the world," "Generally, the more influence China has on other nations, the better off they are," and "The world would be better off if China had greater control over world affairs." (Hyun & Kim, 2015). Responses were rated on 7-point Likert-type scales, ranging from 1 (strongly disagree) to 7 (strongly agree), and then aggregated into a nationalism scale ($M = 5.29$, $SD = 1.25$, $\alpha = .89$).

Demographic variables, including participants' sex, age, and educational level were gauged using conventional measurement.

4. Results

4.1. Descriptive results

We first conducted a series of one-way ANOVA to provide an overall description of the eight groups (see Table 1). The results showed that Chinese AI had the highest level of source credibility ($M = 5.43$, $SD = 0.95$), while American humans were least credible ($M = 3.91$, $SD = 1.48$). No differences were found in terms of nationalism and all other control variables across those eight condition groups. However, there was a significant difference in source credibility ($F(7,536) = 10.66$, partial $\eta^2 = 0.12$, $p < .001$), and message credibility ($F(7,536) = 9.42$, partial $\eta^2 = 0.11$, $p < .001$) among them.

4.2. Hypothesis testing

To test H1-2 and to answer RQ1, we conducted the multiple regression analysis for six condition groups ($n = 417$), including four types of a solo source (domestic human, domestic AI, foreign human, foreign AI) and two groups of a hybrid source with consistent nationality (domestic human-domestic AI, foreign human-foreign AI) (see Table 2). With the non-significant effects of sex, education, pre-existing view and perceived relative advancement between Chinese AI and American AI, the results of Column (2) indicated that labeling AI ($B = 0.695$, $SE = 0.154$, $p < .001$) and hybrid source with human and AI from the same nation ($B = 0.453$, $SE = 0.155$, $p < .001$) can significantly increase source credibility. Additionally, AI ($B = 0.396$, $SE = 0.152$, $p < .01$) and hybrid sources ($B = 0.388$, $SE = 0.153$, $p < .05$) could arouse machine heuristic, which further enhanced source credibility significantly ($B = 0.750$, $SE = 0.034$, $p < .001$) (Columns (1) and (3)). Reported in Columns (3) of Table 2, only the AI source still directly boosted source credibility ($B = 0.398$, $SE = 0.105$, $p < .001$) after the mediation effect of the machine heuristic. Similar results were also observed in message credibility in Columns (4) and (5). Thus, H1 and H2 were confirmed.

H3-4 and RQ2 aimed to investigate the credibility differences between source type (human vs. AI vs. hybrid) and source's nationality (domestic vs. foreign). The findings revealed a foreign source's nationality significantly predicted lower levels of source credibility ($B = -1.224$, $SE = 0.197$, $p < .001$). But AI ($B = 0.725$, $SE = 0.294$, $p < .05$) and hybrid sources ($B = 0.714$, $SE = 0.296$, $p < .05$) narrowed the downward trend caused by source's nationality (Fig. 1). Simple slopes analysis supported it: human source was mostly affected by nationality ($B = -1.192$, $SE = 0.217$, $p < .001$). In contrast, the decline in source credibility of foreign AI ($B = -0.530$, $SE = 0.188$, $p < .01$) or hybrid sources ($B = -0.524$, $SE = 0.224$, $p < .05$) was moderate. Notably, this interaction effect rendered the main effect of source type on source credibility insignificant. Similar interaction effects of nationality and AI ($B = 0.866$, $SE = 0.311$, $p < .01$) or hybrid source ($B = 0.740$, $SE = 0.313$, $p < .05$) were found in message credibility. In terms of message credibility, human ($B = -1.299$, $SE = 0.228$, $p < .001$) was likewise more affected by its nationality than AI ($B = -0.471$, $SE = 0.203$, $p < .05$).

Table 1
Demographics and control variables in eight condition groups.

	US AI	US human	CHN AI	CHN human	US human - US AI	US human - CHN AI	CHN human - US AI	CHN human - CHN AI
Sex ($p = .376$)								
Male	34 (55.74 %)	46 (56.1 %)	44 (63.77 %)	54 (69.23 %)	44 (64.71 %)	29 (50 %)	44 (63.77 %)	36 (61.02 %)
Female	27 (44.26 %)	36 (43.9 %)	25 (36.23 %)	24 (30.77 %)	24 (35.29 %)	29 (50 %)	25 (36.23 %)	23 (38.98 %)
Age ($p = .201$)								
<25	17 (27.87 %)	23 (28.05 %)	28 (40.58 %)	23 (29.49 %)	25 (36.76 %)	18 (31.09 %)	27 (39.13 %)	20 (33.9 %)
25-30	33 (54.1 %)	50 (60.98 %)	40 (57.97 %)	45 (57.69 %)	34 (50 %)	35 (60.34 %)	39 (56.52 %)	33 (55.93 %)
>30	11 (18.03 %)	9 (10.98 %)	1 (1.45 %)	10 (12.82 %)	9 (13.24 %)	5 (8.62 %)	3 (4.35 %)	6 (10.17 %)
Education ($p = .924$)								
High school	14 (22.95 %)	15 (18.29 %)	11 (15.94 %)	22 (28.21 %)	18 (26.47 %)	15 (25.86 %)	14 (20.29 %)	12 (20.34 %)
Bachelor	40 (65.57 %)	54 (65.85 %)	48 (69.57 %)	47 (60.26 %)	41 (60.29 %)	33 (56.9 %)	43 (62.32 %)	37 (62.71 %)
Master and higher	7 (11.48 %)	13 (15.85 %)	10 (14.49 %)	9 (11.54 %)	9 (13.24 %)	10 (17.24 %)	12 (17.39 %)	10 (16.95 %)
Nationalism ($p = .581$)	5.56 (1.14)	5.35 (1.25)	5.30 (1.13)	5.38 (1.39)	5.12 (1.34)	5.17 (1.30)	5.22 (1.21)	5.22 (1.18)
Pre-existing View ($p = .687$)	1.92 (1.32)	2.18 (1.44)	2.33 (1.35)	2.31 (1.51)	2.22 (1.56)	1.97 (1.43)	2.22 (1.59)	2.24 (1.42)
Source Credibility ($p < .001$)	4.97 (1.25)	3.91 (1.48)	5.43 (0.95)	5.16 (1.40)	4.65 (1.32)	4.27 (1.59)	4.67 (1.36)	5.20 (1.18)
Message Credibility ($p < .001$)	5.01 (1.33)	4.02 (1.58)	5.41 (1.07)	5.34 (1.36)	4.67 (1.44)	4.33 (1.64)	4.67 (1.41)	5.27 (1.21)

Notes: 1) The numbers for demographics were frequencies; 2) The numbers for other variables were the means (SD in parentheses).

Table 2
Results of mediation analyses (n = 417).

	Dependent variable:				
	Machine heuristic	Source credibility		Message credibility	
	(1)	(2)	(3)	(4)	(5)
Reference: Source type = Human					
Source type = AI	0.396** (0.152)	0.695*** (0.154)	0.398*** (0.105)	0.551*** (0.163)	0.249* (0.116)
Source type = Hybrid_same	0.388* (0.153)	0.453** (0.155)	0.161 (0.106)	0.341* (0.164)	0.046 (0.117)
Machine heuristic			0.750*** (0.034)		0.760*** (0.037)
Nationalism	0.352*** (0.055)	0.317*** (0.056)	0.054 (0.040)	0.314*** (0.059)	0.047 (0.044)
Other Controls	Yes	Yes	Yes	Yes	Yes
R ²	0.114	0.126	0.604	0.098	0.552
Adjusted R ²	0.099	0.112	0.596	0.083	0.543
Residual Std. Error	1.285 (df = 409)	1.304 (df = 409)	0.879 (df = 408)	1.377 (df = 409)	0.972 (df = 408)
F Statistic	7.522*** (df = 7; 409)	8.460*** (df = 7; 409)	77.675*** (df = 8; 408)	6.380*** (df = 7; 409)	62.856*** (df = 8; 408)

Note: Hybrid_same means hybrid source with human and AI from the same nation. Other controls include sex, education, pre-existing view and perceived relative advancement. *p < .05; **p < .01; ***p < .001.

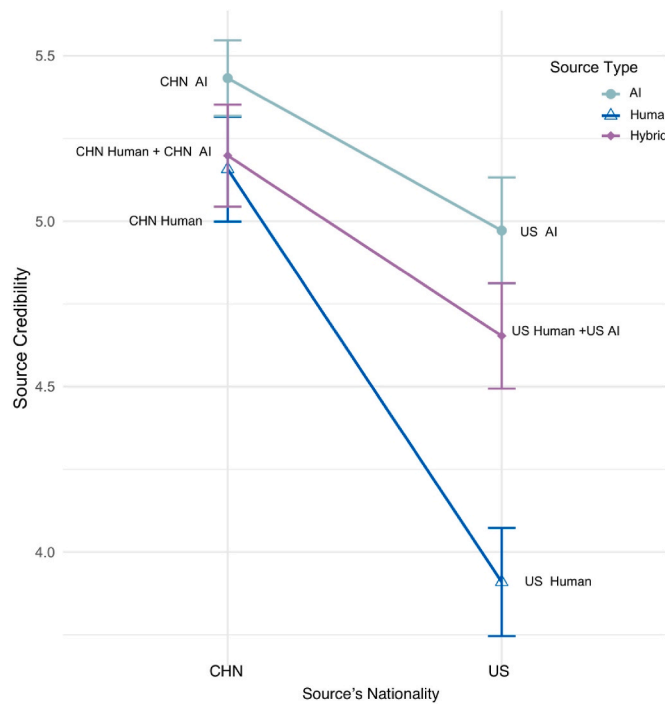


Fig. 1. Interaction effects of source type and source's nationality.

.05) and hybrid sources ($B = -0.576$, $SE = 0.236$, $p < .05$). Thus, H3 and H4 were supported.

To answer RQ3, we analyzed four groups of hybrid sources (foreign human-foreign AI, domestic human-domestic AI, domestic human-foreign AI, and foreign human-domestic AI) ($n = 254$). These four groups can be viewed as consisting of two factors: human's nationality (domestic vs. foreign) and AI's nationality (domestic vs. foreign), thereby investigating the impacts of both factors in the context of hybrid sources. The results showed that in human-AI hybrid sources, both human's foreign nationality ($B = -0.921$, $SE = 0.253$, $p < .001$) and AI's foreign nationality ($B = -0.521$, $SE = 0.242$, $p < .05$) had a significant negative impact on source credibility. A significant interaction effect was found between the AI's nationality and the human's nationality ($B = 0.921$, $SE = 0.344$, $p < .01$). As shown in Fig. 2, to ensure high credibility, both domestic humans and foreign humans should

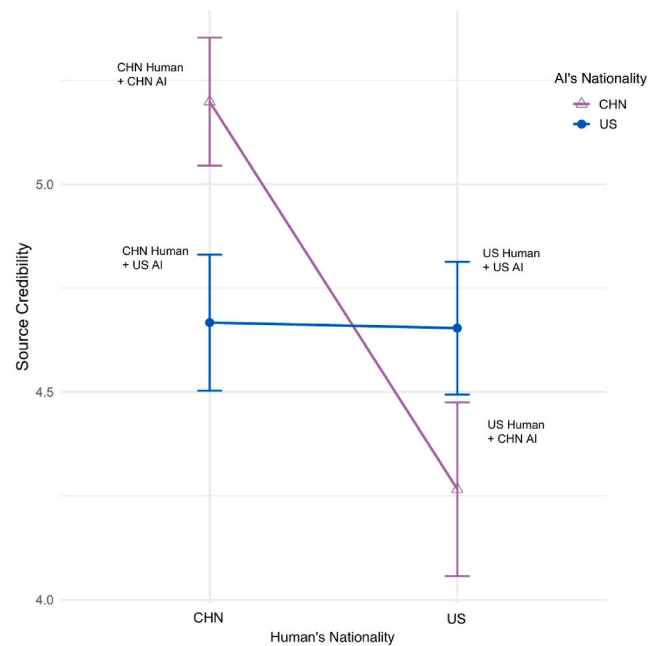


Fig. 2. Interaction effects between human's nationality and AI's nationality in case of hybrid sources.

collaborate with AI from their respective countries. The same main effect of human's nationality ($B = -0.921$, $SE = 0.265$, $p < .001$), main effect of AI's nationality ($B = -0.585$, $SE = 0.253$, $p < .05$) and interaction effect ($B = 0.920$, $SE = 0.360$, $p < .05$) held true for message credibility.

To provide a panoramic view, we finally examined the credibility of the eight source types with the consideration of mediation and moderation (See Table 3, Figs. 4 and 5). The results indicated that domestic sources had higher source credibility ($B = -0.848$, $SE = 0.124$, $p < .001$) and message credibility ($B = -0.922$, $SE = 0.141$, $p < .001$) than those of foreign sources (Columns (3) and (5) in Table 3). AI sources and hybrid sources with human and AI from the same country could not affect the credibility directly. But as mentioned earlier, two approaches existed: the mediation of machine heuristic and the moderation of source's nationality. In contrast, hybrid source with different human and AI nationalities demonstrated a specific way: it did not activate machine heuristic. As shown in Fig. 4, hybrid sources of different human and AI

Table 3
The mediation and moderation effects in eight groups ($N = 544$).

	Dependent variable:				
	Machine heuristic	Source credibility		Message credibility	
	(1)	(2)	(3)	(4)	(5)
Reference: Source type = Human					
Source type = AI	0.397* (0.157)	0.296 (0.215)	0.128 (0.129)	0.085 (0.226)	−0.084 (0.146)
Source type = Hybrid_same	0.375* (0.159)	0.090 (0.224)	−0.171 (0.134)	−0.035 (0.236)	−0.297 (0.153)
Source type = Hybrid_diff	−0.023 (0.159)	−0.442* (0.215)	−0.341** (0.129)	−0.630** (0.226)	−0.529*** (0.146)
Nationality = Foreign		−1.229*** (0.206)	−0.848*** (0.124)	−1.305*** (0.217)	−0.922*** (0.141)
Machine heuristic			0.785*** (0.025)		0.787*** (0.029)
Nationalism	0.283*** (0.050)	0.274*** (0.049)	0.053 (0.030)	0.262*** (0.051)	0.041 (0.034)
AI x Foreign		0.725* (0.308)	0.458* (0.184)	0.860** (0.324)	0.591** (0.209)
Hybrid_same x Foreign		0.713* (0.310)	0.635*** (0.185)	0.738* (0.326)	0.659** (0.211)
Hybrid_diff x Foreign		0.864** (0.310)	0.735*** (0.185)	1.000** (0.326)	0.870*** (0.211)
Other controls	Yes	Yes	Yes	Yes	Yes
R ²	0.083	0.177	0.706	0.154	0.648
Adjusted R ²	0.069	0.159	0.699	0.135	0.639
Residual Std. Error	1.332 (df = 535)	1.296 (df = 531)	0.775 (df = 530)	1.364 (df = 531)	0.881 (df = 530)
F Statistic	6.032*** (df = 8; 535)	9.529*** (df = 12; 531)	98.011*** (df = 13; 530)	8.051*** (df = 12; 531)	75.015*** (df = 13; 530)

Note: Hybrid_same means hybrid source with human and AI from the same nation. Hybrid_diff means hybrid source with human and AI from the different nations. Other controls include sex, education, pre-existing view and perceived relative advancement. * $p < .05$; ** $p < .01$; *** $p < .001$.

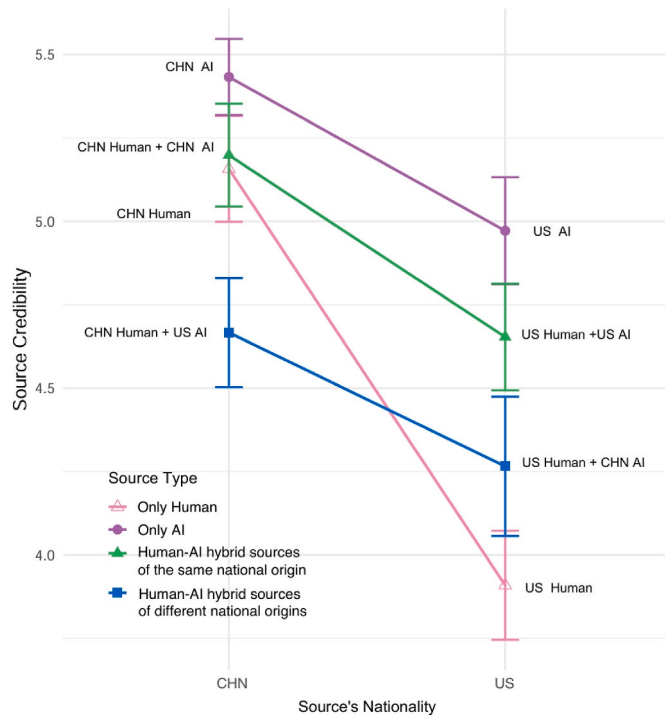


Fig. 3. Differences in source credibility of single sources and hybrid sources.

nationalities directly reduced source credibility ($B = -0.341$, $SE = 0.129$, $p < .01$) and message credibility ($B = -0.529$, $SE = 0.146$, $p < .001$), even if the moderation effect implied that it could likewise mitigate the influence of source's nationality (see Fig. 3).

5. Discussion

5.1. Summary of findings

This study examined the perceived credibility shaped by different authorship with different nationalities. The findings suggested that AI is perceived as more credible than human source, mediated by machine heuristic. For hybrid sources, users assessed credibility based on both human's and AI's nationality. Hybrid sources in which human and AI were originated from the same country could receive more credibility, also through the mediation of machine heuristic. But the credibility will descend vertically when it came to hybrid sources with different human's and AI's nationalities. Furthermore, the interaction effect indicated that the source's nationality greatly influences the credibility of solo human sources, but solo AI and hybrid sources are less affected.

No technology operates in vacuum, but prior studies largely ignore this fact (Moon et al., 2023; Wang & Huang, 2024). This study highlights the significance of those societal and political settings associated with technology: AI from one's own country is more credible than AI from other countries, and users are more likely to trust AI from the same country as humans in the case of hybrid sources. Previous research has pointed out that modern communication technologies reshaped national boundaries while removing them simultaneously (Siddiqi, 2010). Grand national discourses and mass media actively develop nationalism and construct people's perceptions of the political dimensions of technology. For instance, the media usually associates national scientists and technological inventions with national pride (Bucchi et al., 2012). When the state introduces foreign technology, it needs to emphasize the economic benefits to reshape political legitimacy (Dumitrica, 2015). Our study provides empirical evidence from an individual's perspective: Users perceived AI made in their own country as somehow a better source of political communication, which was expected to be used by domestic humans. This suggests that individuals are also actively adapting to the social construction of technology by associating AI sources with national political identity.

Human-AI hybrid sources have been less explored by previous

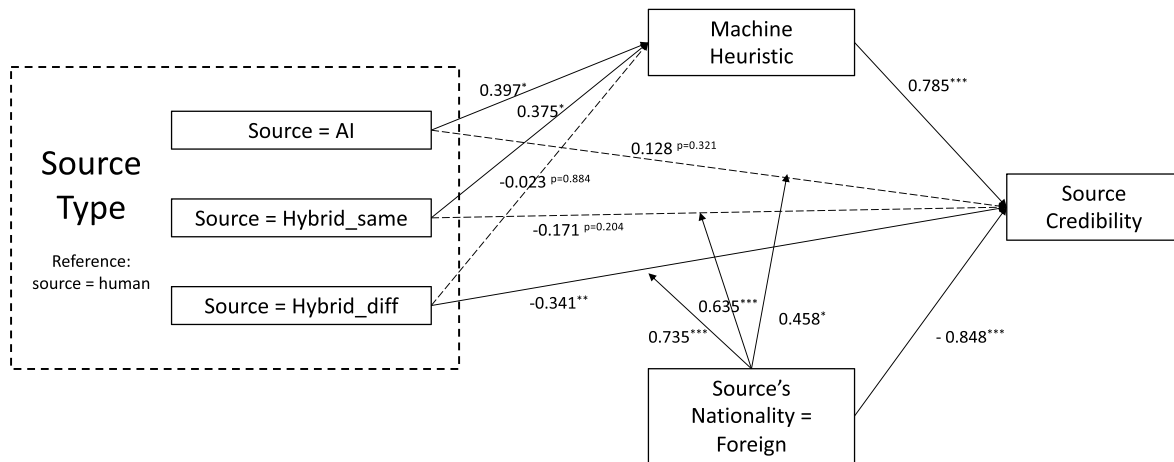


Fig. 4. Mediation and moderation effects on source credibility. Notes: * $p < .05$; ** $p < .01$; *** $p < .001$; the dash lines represent nonsignificant relationships.

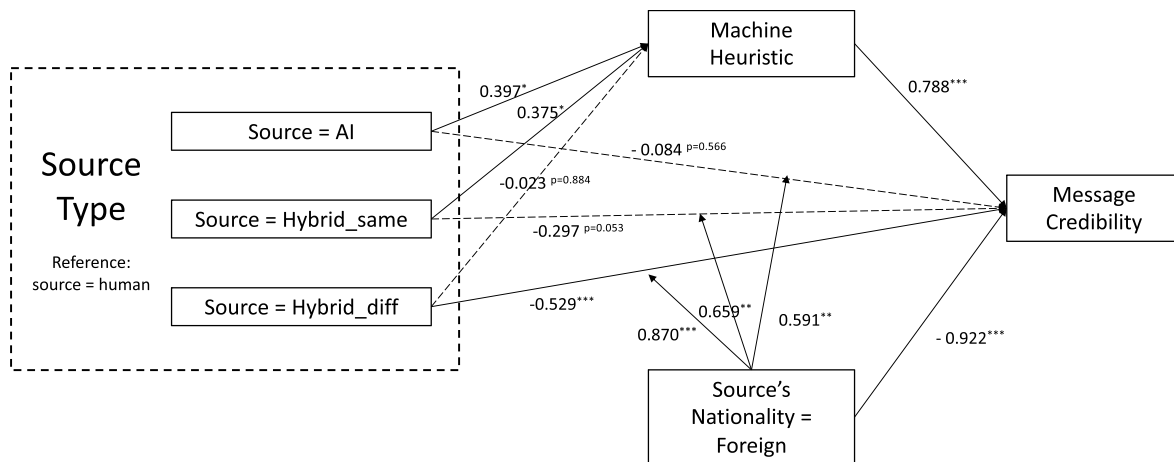


Fig. 5. Mediation and moderation effects on message credibility. Notes: * $p < .05$; ** $p < .01$; *** $p < .001$; the dash lines represent nonsignificant relationships.

research but may represent a future trend. Our findings suggested that AI's nationality matters, as it has a direct impact on credibility. Meanwhile, users will invest more trust in hybrid sources where the AI's nationality matches the human's nationality. When fellow citizens use technologies from other countries, they may be perceived as "betraying their own country" (Zhang, 2022).

Similar to the gendered division of labor in technology (e.g., microwave ovens are perceived as a female technology while computers are a male technology (Rachel, 1995)), in our study, AI technologies are categorized based on the user's nationality: Chinese AI is seen as belonging to Chinese people, while American AI belongs to Americans. Contradicting this expectation of national categorization would trigger a decline in credibility. This progress emphasizes the significance of politics behind new media and technology.

The perception of the political dimension of AI supports the CASA paradigm and media equation theory. Users may mindlessly infer that AI possesses the same political position as humans. It indicates a fundamental lack of trust in external groups as providers of political information, which also extends to AI's credibility. Interestingly, previous research indicated that the human element plays a decisive role in affecting credibility in hybrid sources (Liu & Wei, 2019; Tandoc et al., 2020). Our study not only confirms it, but also reveals the role of AI in hybrid sources. Human-AI hybrid sources are less influenced by human's nationality, which is different from solo human sources. It indicates that the mere presence of AI triggers audience expectations and can somewhat reduce the impact of national origin on credibility. This result

contradicts with prior findings by expanding to the presence of the machine heuristic in hybrid sources, not just in pure machine sources (Cloudy et al., 2022; Sundar & Kim, 2019). It indicates the belief of AI's innate objectivity exists, even if AI is only part of a hybrid source.

The examination of the mediating effects of machine heuristic found that hybrid sources with matching human-AI nationalities exhibit the same mechanisms as solo AI sources. Their facilitating effect on credibility was generated exclusively through the activation of machine heuristics, but machine heuristics will not awaken if someone co-operates with AI from a different country. One potential explanation is that users are engaged in a confrontation between the machine heuristic, which promotes trust in AI, and the political endorsement heuristic, which fosters distrust in foreign sources (Metzger et al., 2010). If human and AI nationalities differ, the user may experience cognitive dissonance and focus more on the nationality issue when evaluating political information. They will access credibility based on the rule that 'people I like usually have correct opinions on issues' [Byrne, 1997, p.4]. As a result, the machine heuristic will be weaker than the political endorsement heuristic under this circumstance. A human-AI hybrid with different nationalities leads to a loss of credibility.

5.2. Theoretical and practical implications

This study expands the scope of research into the machine heuristic. Despite the common perception of machines as objective entities, the machine heuristic has been demonstrated to facilitate the reduction of

motivated reasoning. Nevertheless, this is not a definitive conclusion. The interaction of nationality cues with AI cues alters the impact of the machine heuristic. Nationality cues can hinder AI-derived cues from enhancing an entity's perceived credibility. Although AI is used trans-nationally, the technology can be recognized as having a nationality and making a vital impact [e.g. Ministry of Justice of the, 2022; Waddell, 2018]. This also provides indirect support for the CASA paradigm and media equation theory, which posit that people treat AIs as if they were human and that foreign AIs are perceived as less credible than domestic ones.

This study further explores the dynamics of human-AI hybrid sources. The credibility of a hybrid source has shown mixed results, as it may incorporate both the strengths and weaknesses of both humans and machines (Waddell, 2019). This study finds that the characteristics of AI within hybrid sources also influence outcomes in political information. When AI in a hybrid source shares nationality with the human, it activates machine heuristics and enhances credibility. Conversely, AI with a different nationality from the human in a hybrid source directly reduces credibility. Therefore, the AI's ability to leverage its strengths and trigger machine heuristics to reduce perceived bias depends on its identity cues. Different primary or secondary cues may elicit different levels of social response (Lombard & Xu, 2021). Moreover, accompanied by further integration of automated technology and human sources rather than replacement (Van Dalen, 2012), hybrid sources are not merely a simple combination of humans and AI; the consistency of their identity cues is crucial.

The findings of this study can provide practical guidance for the AIGC field, such as in the context of AI-generated news. On the one hand, it can be observed AI sources have the potential to reduce political bias more effectively than human sources. In the future, AIGC could be introduced into political reporting to help reduce racial discrimination and division, thereby promoting global peace. On the other hand, when utilizing AI in isolation or in conjunction with human input for the generation of content, it is advisable to employ local AI to gain more trust from citizens. This approach is particularly suitable for international organizations or companies seeking to build trust locally. Furthermore, it also underscores the importance of national efforts to achieve technological autonomy.

5.3. Limitations and future research

The findings of this study need to be considered with its limitations. First, the experiments in this study were conducted in China, so replication in other societies needs to consider potential cultural differences. Chinese people generally exhibit higher levels of nationalism, national trust, satisfaction with the government and collectivism than those in many other cultures (Campbell et al., 2021), and they tend to view AIGC more positively than individuals from other cultures (Wu et al., 2020). In particular, future studies should directly compare audiences from different countries, such as China and the United States. A cross-cultural comparative design could help reveal how national context moderates the impact of source nationality, types and heuristic thinking on the perceived credibility of AIGC.

Meanwhile, this study recruited participants from a university in East China, which limits its external validity. A large-scale survey in 41 countries shows a strong correlation between education and a greater cross-national appreciation of science (Noy & O'Brien, 2019). Well-educated samples may have more advanced Internet skills (Wang et al., 2018) and a better understanding of AI (Kim & Lee, 2024). Additionally, past research indicates that college students, as a future elite segment of society, may have lower trust in domestic media and a more open attitude towards foreign sources (Xiang & Hmielowski, 2016; Xu, 2013). Thus, a nationally representative sample is likely to exhibit stronger nationalism, higher trust in domestic sources, a lower understanding of AI, and stronger associations between technology and nationality compared to this study's participants. As a result, we believe

that while sample bias is one of the limitations, it will not overturn the study's findings entirely. Further research can recruit more demographically and regionally diverse participants in China to improve external validity and explore people's political perceptions of AI's nationality in other countries.

To capture the credibility differences brought about by the sources, we selected the same textual story for our online experiment. However, there are differences in the style and language of humans, machines, and hybrid sources until now, because of the AI's limited ability to reason flexibly and understand context (Victor, 2024). In addition, our study excluded samples that failed in the manipulation check. Although they were not significantly different from those in the final sample in terms of the level of nationalism and other control variables, it is a fact that people are cognitive misers and tend to consume content simply without focusing on the source (Fiske & Taylor, 1991). These factors indicate that our experimental setting may not fully replicate real-world scenarios.

In addition, this study was developed partly based on the concept of machine heuristic. However, due to the lack of well-established measurements, we followed a relatively common approach to operationalize it. Research has argued that it is not robust to rely solely on self-reporting to measure the heuristics used by participants (Bellur & Sundar, 2014). This study also ignores some potential confounding variables, such as the context of use and specific biases inherent in the AI models themselves. Although the randomization of the experiment could potentially rule out their influence, the differences in individuals' subjective perceptions may influence credibility. Future research could further explore the operationalization of the machine heuristic and its effect on credibility (Lee, 2023).

Finally, we acknowledge that one limitation of this study is the absence of formal pre-registration. Pre-registration helps enhance the transparency and credibility of scientific findings. Although we developed a detailed research plan in advance and adhered closely to it throughout the study, we recognize that formally pre-registering the study on a public platform would have provided a more standardized and verifiable record of our research intentions and procedures.

6. Conclusion

Responding to the recent crisis of politicization within and beyond science and technology (Druckman, 2017), this paper contributes mainly in two ways. First, this study further explores the credibility of hybrid sources, which is helpful in better grasping the future of human-AI collaboration. The credibility of the hybrid source depends on the human's nationality, the AI's nationality, and the consistency between those two. This paper presents improved cross-national communication practices for content producers and journalists. Second, it provides quantitative empirical evidence about the imagination of socio-political dimensions of technology. Our findings suggested that people's perceptions of technology are not totally neutral. Technologies cannot be understood in isolation from the realities of society. Their perceptions are constructed by nation, society, media and other actors (Taylor, 2004). In the future, AI will be further applied to automated content generation and communication, and will likewise be embedded in the global media landscape. The level of credibility in AI sources and AIGC is not only about the technology itself, but also about the social context in which AI and humans are embedded.

CRedit authorship contribution statement

Junyi Chen: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Weixi Zeng:** Writing – original draft, Methodology, Investigation. **Yi Mou:** Writing – review & editing, Validation, Project administration, Methodology.

Funding

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.techsoc.2025.103032>.

Data availability

Data will be made available on request.

References

- Appelman, A., & Sundar, S. S. (2016). Measuring message credibility: Construction and validation of an exclusive scale. *Journalism and Mass Communication*, 93, 59–79. <https://doi.org/10.1177/1077699015606057>
- Banas, J. A., Palomares, N. A., Richards, A. S., Keating, D. M., Joyce, N., & Rains, S. A. (2022). When machine and bandwagon heuristics compete: Understanding users' response to conflicting AI and crowdsourced fact-checking. *Human Communication Research*, 48, 430–461. <https://doi.org/10.1093/hcr/hqac010>
- Banks, J., Edwards, A. P., & Westerman, D. (2021). The space between: Nature and machine heuristics in evaluations of organisms, cyborgs, and robots. *Cyberpsychology, Behavior, and Social Networking*, 24, 324–331. <https://doi.org/10.1089/cyber.2020.0165>
- Barceló, J., Sheen, G. C.-H., Tung, H. H., & Wu, W.-C. (2022). Vaccine nationalism among the public: A cross-country experimental evidence of own-country bias towards COVID-19 vaccination. *Social Science & Medicine*, 310, Article 115278. <https://doi.org/10.1016/j.socscimed.2022.115278>
- Bellur, S., & Sundar, S. S. (2014). How can we tell when a heuristic has been used? Design and analysis strategies for capturing the operation of heuristics. *Communication Methods and Measures*, 8, 116–137. <https://doi.org/10.1080/19312458.2014.903390>
- Bessi, A., & Ferrara, E. (2016). *Social bots distort the 2016 US presidential election online discussion, first Monday* 21.
- Brewer, M. B., & Chen, Y.-R. (2007). Where (Who) are collectives in collectivism? Toward conceptual clarification of individualism and collectivism. *Psychological Review*, 114, 133–151. <https://doi.org/10.1037/0033-295X.114.1.133>
- Brown, M. B. (2015). Politicizing science: Conceptions of politics in science and technology studies. *Social Studies of Science*, 45, 3–30. <https://doi.org/10.1177/0306312714556694>
- Bryanov, K., Kliegl, R., Koltsova, O., Miltsov, A., Pashakhin, S., Porshnev, A., Sinyavskaya, Y., Terpilovskii, M., & Vziatyshva, V. (2023). What drives perceptions of foreign news coverage credibility? A cross-national experiment including Kazakhstan, Russia, and Ukraine. *Political Communications*, 40, 115–146. <https://doi.org/10.1080/10584609.2023.2172492>
- Bucchi, M. (2012). Visible scientists, media coverage and national identity: Nobel laureates in the Italian daily press. In B. Schiele, M. Claessens, & S. Shi (Eds.), *Science communication in the world: Practices, theories and trends* (pp. 259–268). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-007-4279-6_17
- Burgoon, J. K. (2015). Expectancy violations theory. In *The international encyclopedia of interpersonal communication* (pp. 1–9). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118540190.wbeic102>
- Burton, J. (2023). Algorithmic extremism? The securitization of artificial intelligence (AI) and its impact on radicalism, polarization and political violence. *Technology in Society*, 75, Article 102262. <https://doi.org/10.1016/j.techsoc.2023.102262>
- Byrne, D. (1997). An overview (and underview) of research and theory within the attraction paradigm. *Journal of Social and Personal Relationships*, 14, 417–431. <https://doi.org/10.1177/0265407597143008>
- Campbell, H. A., & La Pastina, A. C. (2010). How the iPhone became divine: New media, religion and the intertextual circulation of meaning. *New Media & Society*, 12, 1191–1207. <https://doi.org/10.1177/1461444810362204>
- Campbell, S. W., Zhao, F., Frith, J., & Liang, F. (2021). Imagining 5G: Public sense-making through advertising in China and the US. *Mobile Media & Communication*, 9, 546–562. <https://doi.org/10.1177/2050157920985239>
- Chaiken, S. (1987). The heuristic model of persuasion. In *Social influence: The Ontario symposium*, 5 pp. 3–39. Hillsdale, NJ, US: Lawrence Erlbaum Associates, Inc.
- Chen, S., & Chaiken, S. (1999). The heuristic-systematic model in its broader context. In *Dual-process theories in social psychology* (pp. 73–96). New York, NY, US: The Guilford Press.
- Chen, K., Shao, A., Burapachee, J., & Li, Y. (2024). Conversational AI and equity through assessing GPT-3's communication with diverse social groups on contentious topics. *Scientific Reports*, 14, 1561. <https://doi.org/10.1038/s41598-024-51969-w>
- Chung, M., Moon, W.-K., & Jones-Jang, S. M. (2023). AI as an apolitical referee: Using alternative sources to decrease partisan biases in the processing of fact-checking messages. *Digital Journalism*, 1–22. <https://doi.org/10.1080/21670811.2023.2254820>
- Cloudy, J., Banks, J., & Bowman, N. D. (2021). The Str(AI)ght scoop: Artificial intelligence cues reduce perceptions of hostile media bias. *Digital Journalism*, 1–20. <https://doi.org/10.1080/21670811.2021.1969974>
- Cloudy, J., Banks, J., & Bowman, N. D. (2022). AI journalists and reduction of perceived hostile media bias: Replication and extension considering news organization cues. *Technology, Mind, and Behavior*. <https://doi.org/10.1037/tmb0000083>
- Cook, G. I., Marsh, R. L., & Hicks, J. L. (2006). Source memory in the absence of successful cued recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32, 828. <https://doi.org/10.1037/0278-7393.32.4.828>
- Diakopoulos, N. (2019). *Automating the news: How algorithms are rewriting the media*. Harvard University Press.
- Dragojlovic, N. (2015). Listening to outsiders: The impact of messenger nationality on transnational persuasion in the United States. *International Studies Quarterly*, 59, 73–85. <https://doi.org/10.1111/isqu.12179>
- Druckman, J. N. (2017). The crisis of politicization within and beyond science. *Nature Human Behaviour*, 1, 615–617. <https://doi.org/10.1038/s41562-017-0183-5>
- Dumitrica, D. (2015). Imagining the Canadian internet: A case of discursive nationalization of technology. *Studies in Ethnicity and Nationalism*, 15, 448–473. <https://doi.org/10.1111/sena.12152>
- Faul, F., Erdfelder, E., Buchner, A., & Lang, A.-G. (2009). Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods*, 41, 1149–1160. <https://doi.org/10.3758/BRM.41.4.1149>
- Feijóo, C., Kwon, Y., Bauer, J. M., Bohlin, E., Howell, B., Jain, R., Potgieter, P., Vu, K., Whalley, J., & Xia, J. (2020). Harnessing artificial intelligence (AI) to increase wellbeing for all: The case for a new technology diplomacy. *Telecommunications Policy*, 44, Article 101988. <https://doi.org/10.1016/j.telpol.2020.101988>
- Fiske, S. T., & Taylor, S. E. (1991). *Social cognition* (2nd ed.). New York: McGraw-Hill.
- Golan, G. J., & Kioussis, S. K. (2010). Religion, media credibility, and support for democracy in the Arab world. *Journal of Media and Religion*, 9, 84–98. <https://doi.org/10.1080/15348421003738793>
- Gollwitzer, A., Martel, C., Brady, W. J., Pärnamets, P., Freedman, I. G., Knowles, E. D., & Van Bavel, J. J. (2020). Partisan differences in physical distancing are linked to health outcomes during the COVID-19 pandemic. *Nature Human Behaviour*, 4, 1186–1197. <https://doi.org/10.1038/s41562-020-00977-7>
- Graefe, A. (2016). Guide to automated journalism. <https://doi.org/10.7916/D80G3XDJ>
- Guzman, A. L. (2019a). Prioritizing the audience's view of automation in journalism. *Digital Journalism*, 7, 1185–1190. <https://doi.org/10.1080/21670811.2019.1681902>
- Guzman, A. L. (2019b). Voices in and of the machine: Source orientation toward mobile virtual assistants. *Computers in Human Behavior*, 90, 343–350. <https://doi.org/10.1016/j.chb.2018.08.009>
- Hancock, J. T., Naaman, M., & Levy, K. (2020). AI-Mediated communication: Definition, research agenda, and ethical considerations. *Journal of Computer-Mediated Communication*, 25, 89–100. <https://doi.org/10.1093/jcmc/zmz022>
- Hart, P. S., Chinn, S., & Soroka, S. (2020). Politicization and polarization in COVID-19 news coverage. *Science Communication*, 42, 679–697. <https://doi.org/10.1177/1075547020950735>
- Henestrosa, A. L., Greving, H., & Kimmerle, J. (2023). Automated journalism: The effects of AI authorship and evaluative information on the perception of a science journalism article. *Computers in Human Behavior*, 138, Article 107445. <https://doi.org/10.1016/j.chb.2022.107445>
- Hong, J.-W., Chang, H.-C. H., & Tewksbury, D. (2025). Can AI become walter cronkite? Testing the machine heuristic, the hostile media effect, and political news written by artificial intelligence. *Digital Journal*, 13, 845–868. <https://doi.org/10.1080/21670811.2024.2323000>
- Huang, G., & Wang, S. (2023). Is artificial intelligence more persuasive than humans? A meta-analysis. *Journal of Communication*, 73, 552–562. <https://doi.org/10.1093/joc/jqad024>
- Hyun, K. D., & Kim, J. (2015). The role of new media in sustaining the status quo: Online political expression, nationalism, and system support in China, information. *Communications Society*, 18, 766–781. <https://doi.org/10.1080/1369118X.2014.994543>
- Jakesch, M., Hancock, J. T., & Naaman, M. (2023). Human heuristics for AI-generated language are flawed. *Proceedings of the National Academy of Sciences*, 120, Article e2208839120. <https://doi.org/10.1073/pnas.2208839120>
- Jang, W., Chun, J. W., Kim, S., & Kang, Y. W. (2023). The effects of anthropomorphism on how people evaluate algorithm-written news. *Digital Journalism*, 11, 103–124. <https://doi.org/10.1080/21670811.2021.1976064>
- Jung, S., Lee, K. M., & Biocca, F. (2015). Voice control system and multiplatform use: Specialist vs. Generalist? In S. Yamamoto (Ed.), *Human interface and the management of information. Information and knowledge design* (pp. 607–616). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-20612-7_57
- Kaa, H., & Krahmer, E. (2014). Journalist versus news consumer : The perceived credibility of machine written news. <https://www.semanticscholar.org/paper/Journalist-versus-news-consumer-%3A-The-perceived-of-Kaa-Krahmer/2dbd9e8fa5d8918a2d25a65c9acbf3996cb9bd>. (Accessed 21 December 2023).
- Kim, S.-W., & Lee, Y. (2024). Investigation into the influence of socio-cultural factors on attitudes toward artificial intelligence. *Education and Information Technologies*, 29, 9907–9935. <https://doi.org/10.1007/s10639-023-12172-y>
- Kim, M., Lee, H., & Kwak, J. (2020). The changing patterns of China's international standardization in ICT under techno-nationalism: A reflection through 5G standardization. *International Journal of Information Management*, 54, Article 102145. <https://doi.org/10.1016/j.ijinfomgt.2020.102145>

- Lee, E.-J. (2023). Minding the source: Toward an integrative theory of human-machine communication. *Human Communication Research*. <https://doi.org/10.1093/hcr/hqad034>
- Lermann Henestrosa, A., & Kimmerle, J. (2024). The effects of assumed AI vs. human authorship on the perception of a GPT-generated text. *Journal Media*, 5, 1085–1097. <https://doi.org/10.3390/journalmedia5030069>
- Lermann Henestrosa, A., & Kimmerle, J. (2025). "always check important information!" - The role of disclaimers in the perception of AI-generated content. *Computers in Human Behavior: Artificial Human*, 4, Article 100142. <https://doi.org/10.1016/j.chbah.2025.100142>
- Lian, Y., Tang, H., Xiang, M., & Dong, X. (2023). Public attitudes and sentiments toward ChatGPT in China: A text mining analysis based on social media. *Technology in Society*, Article 102442. <https://doi.org/10.1016/j.techsoc.2023.102442>
- Liu, B., & Wei, L. (2018). Reading machine-written news: Effect of machine heuristic and novelty on hostile media perception. In M. Kurosu (Ed.), *Human-Computer interaction. Theories, methods, and human issues* (pp. 307–324). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-91238-7_26
- Liu, B., & Wei, L. (2019). Machine authorship in situ. *Digital Journalism*, 7, 635–657. <https://doi.org/10.1080/21670811.2018.1510740>
- Lombard, M., & Xu, K. (2021). Social responses to media technologies in the 21st century: The media are social actors paradigm. *Human-Machine Communication*, 2, 29–55.
- Lopez, G. (2022). *A smarter robot*. The New York Times. <https://www.nytimes.com/2022/12/08/briefing/ai-chatgpt-openai.html>. (Accessed 15 March 2024).
- Lu, Z. (Ed.). (2024). *The routledge handbook of nationalism in east and southeast Asia*. New York, NY: Routledge.
- Luke, H. (2023). Here's how news organisations are using AI in journalism. *Euronews*. <https://www.euronews.com/next/2023/08/24/robot-reporters-heres-how-news-organisations-are-using-ai-in-journalism>. (Accessed 16 March 2024).
- Luther, T., Duzel, V., Ignée, L., Scharfenberg, E., Severing, L., & Kimmerle, J. (2024). *AI in e-commerce: The impact of an AI's alleged age on consumer perceptions of competence and trust*.
- Luther, T., Kimmerle, J., & Cress, U. (2024). Teaming up with an AI: Exploring human-AI collaboration in a writing scenario with ChatGPT. *AI*, 5, 1357–1376. <https://doi.org/10.3390/ai5030065>
- Mann, M., & Schleifer, C. (2020). Love the science, hate the scientists: Conservative identity protects belief in science and undermines trust in scientists. *Social Forces*, 99, 305–332. <https://doi.org/10.1093/sf/soz156>
- Metzger, M. J., Flanagin, A. J., & Medders, R. B. (2010). Social and heuristic approaches to credibility evaluation online. *Journal of Communication*, 60, 413–439. <https://doi.org/10.1111/j.1460-2466.2010.01488.x>
- Meyer, P. (1988). Defining and measuring credibility of newspapers: Developing an index. *Journalism Quarterly*, 65, 567–574. <https://doi.org/10.1177/107769908806500301>
- Ministry of justice of the people's Republic of China, internet user account information management regulations. https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzbmgz/202305/t20230509_478391.html, (2022)–. (Accessed 1 April 2024).
- Molle, A. (2023). A new wave of "space nationalism" in the United States: Why are evangelicals good with it? *Space Policy*, 63, Article 101523. <https://doi.org/10.1016/j.spacepol.2022.101523>
- Moon, W.-K., Chung, M., Mo, S., & Jones-Jang. (2023). How can we fight partisan biases in the COVID-19 pandemic? AI source labels on fact-checking messages reduce motivated reasoning. *Mass Communication & Society*, 26, 646–670. <https://doi.org/10.1080/15205436.2022.2097926>
- Mummendey, A., Klink, A., & Brown, R. (2001). Nationalism and patriotism: National identification and out-group rejection. *British Journal of Social Psychology*, 40, 159–171. <https://doi.org/10.1348/014466601164740>
- Noy, S., & O'Brien, T. L. (2019). Science for good? The effects of education and national context on perceptions of science. *Public Understanding of Science*, 28, 897–916. <https://doi.org/10.1177/0963662519863575>
- Ou, M., Zheng, H., Zeng, Y., & Hansen, P. (2024). Trust it or not: Understanding users' motivations and strategies for assessing the credibility of AI-generated information. *New Media & Society*. <https://doi.org/10.1177/1461448241293154>
- Rachel, J. (1995). Gender and technology in the making. *Feminist Review*, 49, 125–126. <https://doi.org/10.1057/fr.1995.15>
- Reeves, B., & Nass, C. I. (1996). *The media equation: How people treat computers, television, and new media like real people and places*. New York, NY, US: Cambridge University Press.
- Sætra, H. S. (2020). A shallow defence of a technocracy of artificial intelligence: Examining the political harms of algorithmic governance in the domain of government. *Technology in Society*, 62, Article 101283. <https://doi.org/10.1016/j.techsoc.2020.101283>
- Siddiqi, A. A. (2010). Competing technologies, national(ist) narratives, and universal claims: Toward a global history of space exploration. *Techniques & Culture*, 51, 425–443. <https://www.jstor.org/stable/40647107>. (Accessed 4 March 2024).
- Stanley, M. L., Henne, P., Yang, B. W., & De Brigard, F. (2020). Resistance to position change, motivated reasoning, and polarization. *Political Behavior*, 42, 891–913. <https://doi.org/10.1007/s11109-019-09526-z>
- Sun, L., Wei, M., Sun, Y., Suh, Y. J., Shen, L., & Yang, S. (2023). Smiling women pitching down: Auditing representational and presentational gender biases in image-generative AI. *Journal of Computer-Mediated Communication*, 29. <https://doi.org/10.1093/jcmc/zmad045>. zmad045.
- Sundar, S. S. (2008). *The MAIN model: A heuristic approach to understanding technology effects on credibility*. MA: MacArthur Foundation Digital Media and Learning Initiative Cambridge.
- Sundar, S. S., & Kim, J. (2019). Machine heuristic: When we trust computers more than humans with our personal information. In *Proceedings of the 2019 CHI conference on human factors in computing systems* (pp. 1–9). Glasgow Scotland Uk: ACM. <https://doi.org/10.1145/3290605.3300768>.
- Tajfel, H., & Turner, J. C. (2004). *The social identity theory of intergroup behavior*. In *Political psychology* (pp. 276–293). Psychology Press.
- Tan, S.-M., & Liew, T. W. (2022). Multi-Chatbot or Single-Chatbot? The effects of M-Commerce chatbot interface on source credibility, social presence, trust, and purchase intention. *Human Behavior and Emerging Technologies*, 2022, 1–14. <https://doi.org/10.1155/2022/2501538>
- Tandoc, E. C., Jr., Yao, L. J., & Wu, S. (2020). Man vs. machine? The impact of algorithm authorship on news credibility. *Digital Journalism*, 8, 548–562. <https://doi.org/10.1080/21670811.2020.1762102>
- Tang, M. (2020). Huawei versus the United States? The geopolitics of extraterritorial internet infrastructure. *International Journal of Communication*, 14, 22.
- Tay, B., Jung, Y., & Park, T. (2014). When stereotypes meet robots: The double-edge sword of robot gender and personality in human-robot interaction. *Computers in Human Behavior*, 38, 75–84.
- Taylor, C. (2004). *Modern social imaginaries*. Durham: Duke University Press.
- Van Dalen, A. (2012). The algorithms behind the headlines: How machine-written news redefines the core skills of human journalists. *Journalism Practice*, 6, 648–658. <https://doi.org/10.1080/17512786.2012.667268>
- Verbeek, P.-P. (2017). The struggle for technology: Towards a realistic political theory of technology. *Foundations of Science*, 22, 301–304. <https://doi.org/10.1007/s10699-015-9470-7>
- Victor, B. (2024). The limits of large language models. *Datacenterdynamics*. <https://www.datacenterdynamics.com/en/opinions/the-limits-of-large-language-models/>. (Accessed 16 March 2024).
- Waddell, T. F. (2018). A robot wrote this? *Digital Journalism*, 6, 236–255. <https://doi.org/10.1080/21670811.2017.1384319>
- Waddell, T. F. (2019). Can an algorithm reduce the perceived bias of news? Testing the effect of machine attribution on news readers' evaluations of bias, anthropomorphism, and credibility. *Journalism and Mass Communication*, 96, 82–100. <https://doi.org/10.1177/1077699018815891>
- Wang, H., Cai, T., Mou, Y., & Shi, F. (2018). Traditional resources, internet resources, and youth online political participation: The resource theory revisited in the Chinese context. *Chinese Sociological Review*, 50, 115–136. <https://doi.org/10.1080/21620555.2017.1341813>
- Wang, S., & Huang, G. (2024). The impact of machine authorship on news audience perceptions: A meta-analysis of experimental studies. *Communication Research*, Article 00936502241229794. <https://doi.org/10.1177/00936502241229794>
- Wölker, A., & Powell, T. E. (2021). Algorithms in the newsroom? News readers' perceived credibility and selection of automated journalism. *Journalism*, 22, 86–103. <https://doi.org/10.1177/1464884918757072>
- Wu, Y., Mou, Y., Li, Z., & Xu, K. (2020). Investigating American and Chinese subjects' explicit and implicit perceptions of AI-Generated artistic work. *Computers in Human Behavior*, 104, Article 106186. <https://doi.org/10.1016/j.chb.2019.106186>
- Xiang, J., & Hmielowski, J. D. (2016). Alternative views and eroding support: The conditional indirect effects of foreign media and internet use on regime support in China. *International Journal of Public Opinion Research*, Article edw006. <https://doi.org/10.1093/ijpor/edw006>
- Xu, J. (2013). Trust in Chinese state media: The influence of education, internet, and government. *Journal of International Communication*, 19, 69–84. <https://doi.org/10.1080/13216597.2012.737816>
- Xu, K., Chen, X., & Huang, L. (2022). Deep mind in social responses to technologies: A new approach to explaining the computers are social actors phenomena. *Computers in Human Behavior*, 134, Article 107321. <https://doi.org/10.1016/j.chb.2022.107321>
- Xu, K., Liu, F., Mou, Y., Wu, Y., Zeng, J., & Schäfer, M. S. (2020). Using machine learning to learn machines: A cross-cultural study of users' responses to machine-generated artworks. *Journal of Broadcasting & Electronic Media*, 64, 566–591. <https://doi.org/10.1080/08838151.2020.1835136>
- Zhang, J. (2022). Hardening national boundaries in a globally-connected world: Technology, development and nationalism in China. *Journal of Contemporary Asia*, 52, 783–804. <https://doi.org/10.1080/00472336.2021.2001841>
- Zhang, Z. (2024). Technology and geopolitics: The social construction of Huawei's 5G controversy in Europe. *Global Media and Communication*, 20, 217–235. <https://doi.org/10.1177/17427665241251448>
- Zhu, Y., Bowen, S. A., & Lyu, X. (2022). Messenger nationality, media skepticism, and crisis communication effectiveness: Huawei's YouTube messages as perceived in the U.S. *International Journal of Strategic Communication*, 16, 111–126. <https://doi.org/10.1080/1553118X.2021.2014499>